UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2011 question paper for the guidance of teachers

5014 ENVIRONMENTAL MANAGEMENT

5014/22 Paper 2, maximum raw mark 60

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1 (a) (i) rain gauge; [1]

(ii) labelled axes;; plots;; [4]

(iii) high rainfall associated with low yield/converse/eq;
yields go up after high rainfall;
yields go down after below average rainfall;
ref to figures for two years in context;
(any two) [2]

(iv) 75(%);; [2]

(v) 880; 1.0; [2]

(b) (i) relying on only two crops;
risk of crop failure;
drop in price;
only mango can be used as food by farmer;
(any two) [2]

(ii) More food;
less risk of all crops failing;
some fuel gained;
(any one) [1]

(iii) different crops need different nutrients;
crops can be rotated;
good nutrition;
more value of production;
land very unlikely to be degraded;
green manures possible;
AVP e.g. soil exhaustion;
(any three) [3]

(iv) old trees may give less cloves/die;
new trees take time to be productive/eq;
help to maintain supply of cloves;
(any one) [1]

2 (a) (i) at random/eq; [1]

(ii) snapper;
only smaller fish left/eq; [2]

(iii) experimental error;
immigration of longer/larger/older fish;
faster growth rate in fishing zone;
(any one) [1]

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(iv) same size net;
further detail e.g. size/shape of holes;
fish for same time;
same depth;
AVP e.g. same season;
(any two) [2]

(v) repeat the sampling;
more time/stated time;
repeat around another island; [3]

(b) (i) (1) 4 *3 2 6 5 (7)*
(5 correct 2 marks;; 3/4 correct 1 mark; less than 3 correct no mark.) [2]

(ii) table layout suitable;
suitable titles e.g. fish length (cm)/fish species/snapper/jacks/wrasse (any order);
average length (cm); [3]

(iii) less mature fish left to breed;
so less very small fish to grow in future;
ref to changes in food chain; [3]

3 (a) (i) more tourism/eq; [1]

(ii) collect seed;
make compost of forest leaves/eq;
plant in pots;
water;
shelter from sun/wind;
AVP;
(any four) [4]

(iii) in gaps in 'old forest';
on edges of/in clove plantations to see if they survive;
record height of trees;
how many living trees remain/ numbers of living/dead trees;
(any two) [2]

(b) (i) one road drawn;
three paths;
(must be in sensible locations) [2]

(ii) reasonable improvements such as toilets; accommodation; telephone/internet; food imports for tourists; (any three)
good reason for each ;;; (any three) [6]

(c) (i) three questions, at least two linked to idea of 'tourist suggestions' e.g. more informed guides/interpretation boards/hides for observation/AVP;;;
layout; [4]

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(ii) add up each response option;
find % for each response option/eq; [2]

(d) (i) more mats made for sale to tourists;
more cooking needed for tourists;
more colourful mats get higher prices/eq;
(any two) [2]

(ii) earn more money from mats;
more jobs;
as a cash crop;
AVP;
(any two) [2]

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